JOIN TABLES EXERCISE 4

TABLE 1: EMPLOYEES

EMPLOYEE ID	FIRST NAME	LAST NAME	DEPARTMENT	SALARY	CITY	AGE
1	JOHN	DOE	IT	70000	NEW YORK	35
2	ALICE	SMITH	HR	60000	LOS ANGELES	40
3	BOB	JOHNSON	FINANCE	75000	TORONTO	45
4	DAVID	BROWN	IT	72000	LONDON	30
5	EMMA	WILSON	SALES	65000	SYDNEY	28
6	MICHAEL	CLARK	FINANCE	80000	NEW YORK	50

TABLE 2: PROJECTS

PROJECT ID	EMPLOYEE ID	PROJECT NAME	BUDGET	START DATE	STATUS
101	1	AI DEVELOPMENT	100000	2024-01-10	COMPLETED
102	2	EMPLOYEE TRAINING	50000	2024-02-15	ONGOING
103	1	CYBERSECURITY AUDIT	75000	2024-03-05	PENDING
104	3	FINANCIAL ANALYSIS	90000	2024-04-12	ONGOING
105	5	MARKET EXPANSION	65000	2024-05-20	COMPLETE
106	6	RISK MANAGEMENT	80000	2024-06-10	PENDING

JOIN STATEMENT EXERCISE 4

① RETRIEVE ALL EMPLOYEES AND THEIR ASSIGNED PROJECTS USING AN INNER JOIN
QUERY RETURN: EMPLOYEEID, FIRSTNAME, LASTNAME, DEPARTMENT, SALARY, PROJECTID, PROJECTNAME, BUDGET, STATUS

```
# SELECT A.`EMPLOYEEID`, FIRSTNAME, LASTNAME, DEPARTMENT, SALARY, `PROJECTID`, `PROJECTNAME`,
        BUDGET, STATUS
FROM `WORKSPACE`.`DEFAULT`.`TABLE_1_EMPLOYEES_SQL_JOIN` AS A
INNER JOIN `WORKSPACE`.`DEFAULT`.`TABLE_2_PROJECTS_SQL_JOIN` AS B
ON `A`.`EMPLOYEEID` = `B`.`EMPLOYEEID`;
```

EMPLOYEEID	FIRSTNAME	LASTNAME	DEPARTMENT	SALARY	PROJECTID	PROJECTNAME	BUDGET	STATUS
1	JOHN	DOE	IT	70000	101	AI DEVELOPMENT	100000	COMPLETED
2	ALICE	SMITH	HR	60000	102	EMPLOYEE TRAINING	50000	ONGOING
3	JOHN	DOE	IT	70000	103	CYBERSECURITY AUDIT	75000	PENDING
4	BOB	JOHNSON	FINANCE	75000	104	FINANCIAL ANALYSIS	90000	ONGOING
5	EMMA	WILSON	SALES	65000	105	MARKET EXPANSION	65000	COMPLETED
6	MICHAEL	CLARK	FINANCE	80000	106	RISK MANAGEMENT	80000	PENDING

② RETRIEVE ALL EMPLOYEES AND THEIR ASSIGNED PROJECTS, INCLUDING EMPLOYEES WHO HAVE NO PROJECTS. LEFT JOIN.

```
# SELECT A.`EMPLOYEEID`, FIRSTNAME, LASTNAME, DEPARTMENT, SALARY, `PROJECTID`, `PROJECTNAME`, BUDGET, STATUS
FROM `WORKSPACE`.`DEFAULT`.`TABLE_1_EMPLOYEES_SQL_JOIN` AS A
LEFT JOIN `WORKSPACE`.`DEFAULT`.`TABLE_2_PROJECTS_SQL_JOIN` AS B
ON A.EMPLOYEEID = B.EMPLOYEEID;
```

EMPLOYEEID	FIRSTNAME	LASTNAME	DEPARTMENT	SALARY	PROJECTID	PROJECTNAME	STATUS	BUDGET
1	JOHN	DOE	IT	70000	103	CYBERSECURITY AUDIT	PENDING	75000
2	ALICE	SMITH	HR	60000	102	EMPLOYEE TRAINING	ONGOING	50000
3	BOB	JOHNSON	FINANCE	75000	104	FINANCIAL ANALYSIS	ONGOING	90000
4	DAVID	BROWN	IT	72000	NULL	NULL	NULL	NULL
5	EMMA	WILSON	SALES	65000	105	MARKET EXPANSION	COMPLETED	65000
6	MICHAEL	CLARK	FINANCE	80000	106	RISK MANAGEMENT	PENDING	80000
1	JOHN	DOE	IT	70000	101	AI DEVELOPMENT	COMPLETED	100000

③ Retrieve all projects and their assigned employees, including projects that have no employee. Right Join
Query Return: Project ID, Project Name, Budget, Status, Employee ID, First Name, Last Name, Department, Salary

```
# SELECT B.`Project ID`, Project Name, Budget, Status, A. Employee ID, `First Name`, `Last Name`, Department, Salary
FROM `workspace`.`default`.`table_1_employees_sql_join` AS A
RIGHT JOIN `workspace`.`default`.`table_2_projects_sql_join` AS B
ON A. Employee ID = B. Employee ID;
```

Project ID	Project Name	Budget	Status	Employee ID	First Name	Last Name	Department	Salary
101	AI Development	100000	Completed	1	John	Doe	IT	70000
102	Employee Training	50000	Ongoing	2	Alice	Smith	HR	60 000
103	Cybersecurity Audit	75000	Pending	1	John	Doe	IT	70 000
104	Financial Analysis	90000	Ongoing	3	Bob	Johnson	Finance	75000
105	Market Expansion	65000	Completed	5	Emma	Wilson	Sales	65000
106	Risk Management	80000	Pending	6	Michael	Clark	Finance	80 000

④ Retrieve all employees and projects, including those without a match in either table. Full Outer Join
Query Return: Employee ID, First Name, Last Name, Department, Salary, Project ID, Project Name, Budget, Status

```
# SELECT A.`Employee ID`, First Name, Last Name, Department, Salary, `Project ID`, Project Name, Budget, Status
FROM `workspace`.`default`.`table_1_employees_sql_join` AS A
FULL OUTER JOIN `workspace`.`default`.`table_2_projects_sql_join` AS B
ON A. Employee ID = B. Employee ID;
```

Employee ID	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	null	null	null	null
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
1	John	Doe	IT	70000	101	AI Development	100000	Completed

(5) RETRIEVE A LIST OF ALL UNIQUE CITIES WHERE EMPLOYEES ARE LOCATED AND PROJECT STATUSES

QUERY RETURN: LOCATION (RENAME THE COLUMN TO LOCATION USING AN ALIAS)

```
# SELECT STATUS AS LOCATION
FROM `WORKSPACE`.`DEFAULT`.`TABLE_2_PROJECTS_SQL_JOIN`
UNION
SELECT CITY AS LOCATION
FROM `WORKSPACE`.`DEFAULT`.`TABLE_1_EMPLOYEES_SQL_JOIN`;
```

LOCATION
COMPLETED
ONGOING
PENDING
NEW YORK
LOS ANGELES
TORONTO
LONDON

LOCATION
COMPLETED
ONGOING
PENDING
NEW YORK
LOS ANGELES
TORONTO
LONDON
SYDNEY

(6) RETRIEVE A LIST OF ALL CITIES WHERE EMPLOYEES ARE LOCATED AND PROJECT STATUSES, ALLOWING DUPLICATES

QUERY RETURN: LOCATION (RENAME THE COLUMN TO LOCATION USING AN ALIAS)

```
# SELECT STATUS LOCATION
FROM `WORKSPACE`.`DEFAULT`.`TABLE_2_PROJECTS_SQL_JOIN`
UNION ALL
SELECT CITY AS LOCATION
FROM `WORKSPACE`.`DEFAULT`.`TABLE_1_EMPLOYEES_SQL_JOIN`;
```

LOCATION
COMPLETED
ONGOING
PENDING
ONGOING
COMPLETED
PENDING
NEW YORK
LOS ANGELES
~~LOCATION~~
TORONTO
LONDON
SYDNEY
NEW YORK

7) RETRIEVE EMPLOYEES WHO EARN MORE THAN 70 000

QUERY RETURN: EMPLOYEE ID, FIRST NAME, LAST NAME, DEPARTMENT, SALARY

```
# SELECT EMPLOYEE ID, FIRST NAME, LAST NAME, DEPARTMENT, SALARY
FROM `WORKSPACE`.`DEFAULT`.`TABLE_1_EMPLOYEES_SQL_JOIN`
WHERE SALARY > 70000;
```

EMPLOYEE ID	FIRST NAME	LAST NAME	DEPARTMENT	SALARY
3	BOB	JOHNSON	FINANCE	75000
4	DAVID	BROWN	IT	72000
6	MICHAEL	CLARK	FINANCE	80000

8) RETRIEVE EMPLOYEE WORKING IN EITHER IT OR FINANCE DEPARTMENT

QUERY RETURN: EMPLOYEE ID, FIRST NAME, LAST NAME, DEPARTMENT, SALARY

```
# SELECT EMPLOYEE ID, FIRST NAME, LAST NAME, DEPARTMENT, SALARY
FROM `WORKSPACE`.`DEFAULT`.`TABLE_1_EMPLOYEES_SQL_JOIN
WHERE DEPARTMENT IN ('IT', 'FINANCE');
```

EMPLOYEE ID	FIRST NAME	LAST NAME	DEPARTMENT	SALARY
1	JOHN	DOE	IT	70000
3	BOB	JOHNSON	FINANCE	75000
4	DAVID	BROWN	IT	72000
6	MICHAEL	CLARK	FINANCE	80000

(9) RETRIEVE PROJECTS THAT ARE NOT YET COMPLETED

QUERY RETURN: PROJECT ID, PROJECT NAME, BUDGET, STATUS

```
# SELECT PROJECT ID, PROJECT NAME, BUDGET, STATUS
FROM `WORKSPACE`.`DEFAULT`.`TABLE_2_PROJECTS_SQL_JOIN`
WHERE STATUS != 'COMPLETED';
```

PROJECT ID	PROJECT NAME	BUDGET	STATUS
102	EMPLOYEE TRAINING	50 000	ONGOING
103	CYBERSECURITY AUDIT	75000	PENDING
104	FINANCIAL ANALYSIS	90000	ONGOING
106	RISK MANAGEMENT	80000	PENDING

(10) RETRIEVE PROJECTS THAT HAVE A BUDGET GREATER THAN 70 000 AND ARE NOT COMPLETED

QUERY RETURN: PROJECT ID, PROJECT NAME, BUDGET, STATUS

```
# SELECT PROJECT ID, PROJECT NAME, BUDGET, SALARY
FROM `WORKSPACE`.`DEFAULT`.`TABLE_2_PROJECTS_SQL_JOIN`
WHERE BUDGET > 70 000 AND STATUS != 'COMPLETED';
```

PROJECT ID	PROJECT NAME	BUDGET	STATUS
103	CYBERSECURITY AUDIT	75000	PENDING
104	FINANCIAL ANALYSIS	90000	ONGOING
106	RISK MANAGEMENT	80000	PENDING